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Long-Term Applications of a Wireless Intelligent Safety Valve

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Abstract

The maintenance of well integrity is critical to ensure the safety and longevity of aging assets. Safety systems such as downhole safety valves often require replacement during the life of the well and retrofits often do not provide the same level of safety as the original system. This paper introduces a new technology that utilises wireless command, control, monitoring and autonomy to provide an intelligent safety valve retrofittable in existing completions. The technology has been field deployed in a variety of environments as a reliable replacement for the safety valve. More recently the field application and track record of this technology have proven the feasibility of long-term performance under a variety of downhole conditions. It has also demonstrated substantial cost and CO₂ savings of retrofitting versus recompletion.

The safety valve is designed to be deployed or retrofitted in an existing completion and operate responding to wireless commands from surface or operating autonomously in response to a change in a well's flowing conditions. The valve is configured to respond to absolute and dynamic triggers that indicate a loss of containment. Most variants were deployed via slickline intervention with the option to retrieve later in the well life and replace when necessary. The intelligent safety valve is an autonomous, API14A qualified, device that enables the safe production of wells with failed downhole safety valves. This valve was installed in both oil and gas wells to restore production under a variety of challenging production conditions. We will review a wide range of field applications of this safety valve, demonstrate a significant run history and discuss the challenges overcome through application.

The production of multiple wells was restored by retrofitting the wireless intelligent safety valve over the last four years. The valve successfully operated under a variety of challenging conditions including cyclic loading, gas lift, velocity string and slug production. The retrofit ability of this technology significantly reduced environmental impact and overall project costs.

This is the first publication that demonstrates the long-term performance of this technology in a variety of applications under downhole conditions. This is the only wireless control and command safety valve that does not rely on repeaters or additional surface equipment. This is the only safety valve that transmits valve status and health to surface on a daily basis thereby confirming tool function and well integrity. The valve can be wirelessly surface commanded to shut-in for emergency shutdown events or testing and then reopened without intervention. The valve is capable of operating under a variety of production conditions often encountered in mature assets such as cyclic production and liquid loading.

Introduction

In 2023, the top reported issues with United Kingdom Continental Shelf well stock were related to well integrity, with 31% of well integrity issues related to Down Hole Safety Valves (DHSV), majority of which are shut-in, and production ceased ([North Sea Transition Authority 2024 Well Insights Report](#)). Typically, for TRSCSSV failures a workover is required to replace the valve, equating to approximately ten times the cost of repair or retrofit solutions ([North Sea Transition Authority 2024 Well Insights Report](#)). It is therefore understandable that technology demands have increased to align with intervention-based repair or replacement solutions in UKCS to return shut-in wells to production while avoiding high workover costs.

Wireless Command Control and Monitoring

Wireless downhole communication overcomes the challenge to instrument and add functionality to wells without the use of control lines, feedthroughs and interfaces while enabling the installation in new completions or retrofit into existing completion architecture ([Raw 2007](#), [Naldrett 2012](#)). There are a range of downhole communication platforms available in the industry each with their pros- and cons- extensively reviewed by [Bouldin et al. in 2021](#). Bouldin's paper provides a summary of the state-of-the-art in 2019, yet more recent publications demonstrate that the development and application technology is progressing rapidly ([Abbott 2024](#)).

Wireless command, control, and monitoring technology ([Figure 1](#)) overcomes the scalability challenges associated with addressing a diverse and aging well stock. Removing the complexity of cables and control lines makes this technology simple to deploy with the completion and easily retrofittable into existing completions via straightforward intervention techniques such as slickline or coiled tubing. These devices can be commanded from surface or programmed to operate autonomously and react to the downhole environment without command from surface. If required, surface monitoring or control equipment is then easily integrated with surface infrastructure. System performance is routinely monitored at surface to plan service or timely replacement of these devices and ensure continuity of operations. This technology can often function easily without integration into existing control systems, which can always be done at a later date when required.



Figure 1—Example of fluid flowing from the tool's annulus to tubing, via the choke ports. Controlled manipulation of the choke generates pressure signals to surface.

Wireless command, control and monitoring technology is mainly limited by the available power which today is based on lithium battery technology. The working life of these devices is dictated by the power consumption or usage and available battery life. Although power generation solutions such as turbines exist for downhole tools, there is currently no reliable rechargeable battery technology for long-term application at high temperatures. The industry continues research and development to both reduce power usage of these devices while extending the downhole longevity of battery technology.

These systems provide real-time monitoring of downhole parameters and wirelessly relay this data to surface for reservoir and production optimization or well integrity monitoring. Monitoring is often combined with control to enable remote control of flow at desired locations in the upper or lower completion. This technology continues to be applied in a variety of downhole applications: remote monitoring for well test, well test valve actuation, replace failed permanent downhole gauges ([Green 2018](#)), adding flow control and

monitoring to existing wells, multiple devices for control of flow contribution from segmented production intervals, replace a failed sub-surface safety valve and downhole barrier monitoring ([Azevedo 2023](#)). The next section discusses the application of this technology for remediation of failed safety valves.

Intelligent Safety Valve

The wireless Intelligent Safety Valve (ISV) is a retrofittable, battery operated, electro-mechanical sub-surface safety valve. The electro-mechanical design allows for a wide operating window, combining dynamic and absolute control measures for responding to a loss of surface containment (LoSC) and emergency shut-down (ESD) safety events. Communication to and from the tool is a patent protected method, utilizing pressure modulation for data transmission and command.

The ISV can be deployed on a lock, flow through plug or retrievable packer using conventional intervention methods. It is equipped with quartz pressure gauges for monitoring pressure to recognize emergency scenarios and to receive signature pulses from surface in order to re-open the valve. The tool is an API 14A qualified device, that can be specifically configured to perform in a range of well conditions.

The valve is designed to recognize and respond to dynamic pressure conditions. Equipped with four pre-defined closure trigger scenarios, it offers a wide range of options and assurance for emergency event detection combining both dynamic and absolute methods. The valve can be configured to ignore sporadic pressure fluctuations that are out with normal flowing conditions. It has the capability of applying a more intelligent approach; preventing inadvertent closures while ensuring accurate surveillance and action when it identifies any of its activated closure scenarios. The tool isolates 95% of flow in 8 seconds and is fully closed and sealed in 22 seconds. Following ISV closure, when the well is in a no flow scenario, a unique, configurable pressure bleed down or pressure build up protocol are followed to initiate an open command and reopen the tool. The valve can be re-opened with a differential up to 1,500psi, removing the need to fully equalize.

ISV closure in an ESD scenario is triggered when the stable flowing pressure, as monitored by the valve, increases by a pre-configured magnitude within a pre-configured time ([Figure 2](#)). The pressure and time values to which the ESD trigger is configured are specific to the well and are programmed to the tool prior to installation. In a similar way, during a LoSC scenario, the valve is triggered when it detects the flowing pressure drop at a rate that exceeds normal flowing conditions. The magnitude of pressure drop and the time over which this occurs are again specific to the individual well conditions. This dynamic method of triggering the ISV to a closed position allows for the steady depletion of stable flowing pressures over time without the risk of inadvertently closing the valve when the pressure reaches a fixed absolute value. Unlike traditional velocity type valves, the ISV will adjust the boundaries of its safe working pressure window to account for reservoir depletion and, as a result, can remain active in hole for a much greater duration without the requirement for resetting or recalibrating for well conditions. As can be seen in this paper, the limiting factor in the life of an ISV in hole becomes battery duration rather than loss of functionality.

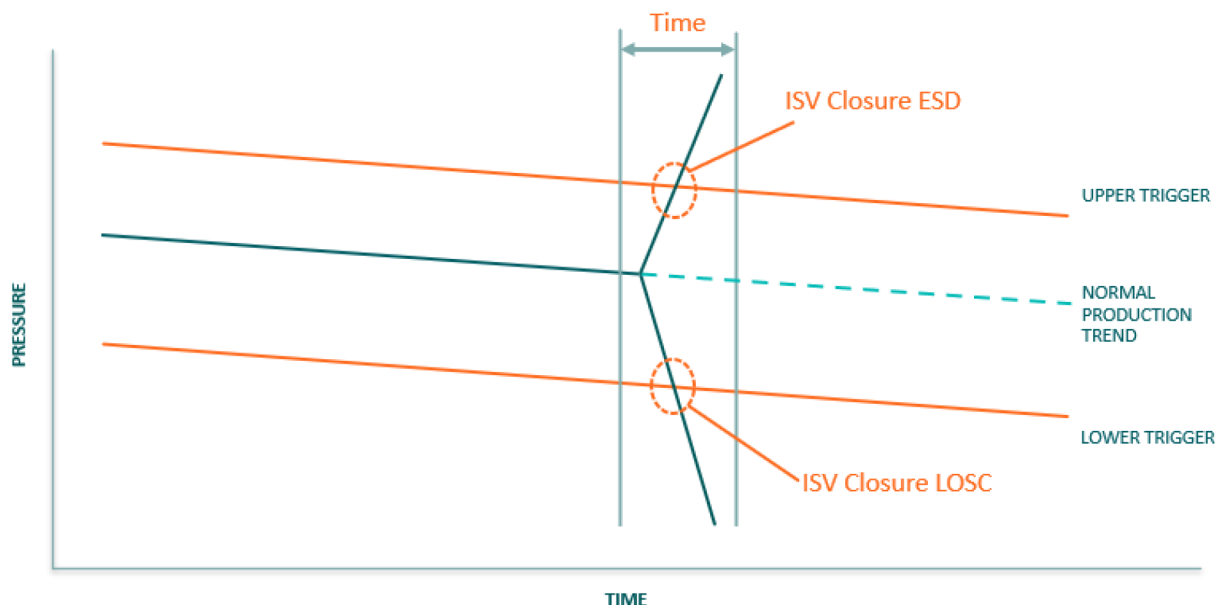


Figure 2—Programmable dynamic triggers for Loss of Surface Containment (LoSC) and Emergency Shutdown (ESD) safety events. Pre-programmed to align with well dynamics, these triggers operate using a rate of change of pressure. An increase or decrease in pressure over time. The configured values are determined by looking at known or predicted Flowing Tubing Head Pressures (FTHP).

In addition to the dynamic triggers, there is also the option of using an absolute, pre-defined value as a lower trigger (Figure 3). This option can be used in addition to the dynamic triggers, or as an alternative single trigger mechanism. The absolute value chosen is programmed to the tool's memory and, unlike traditional valves, is independent of temperature variations at setting depth or significant pressure drift over time.

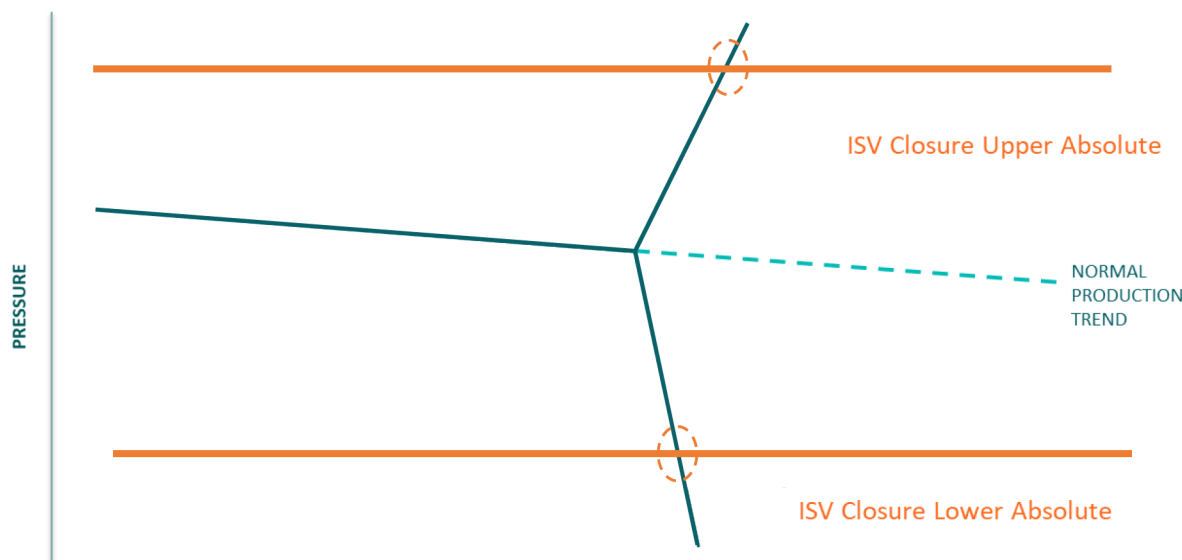


Figure 3—Upper and Lower Absolute pressure values, independent of time.

The ISV is configured to send regular Vitality Pulses (VP) from the tool to surface. The VP provides a functional test of all critical systems and confirms that the valve movement is uninhibited (Figure 5). The delivery time of the Vitality Pulse provides valuable battery life data reported directly from the tool's on-

board monitoring system. Regular reporting of battery voltage ensures adequate time and planning for valve replacement, predicting the remaining time in well to ensure optimum use.



Figure 4—Vitality Pulse detection using existing WHP monitoring sensors

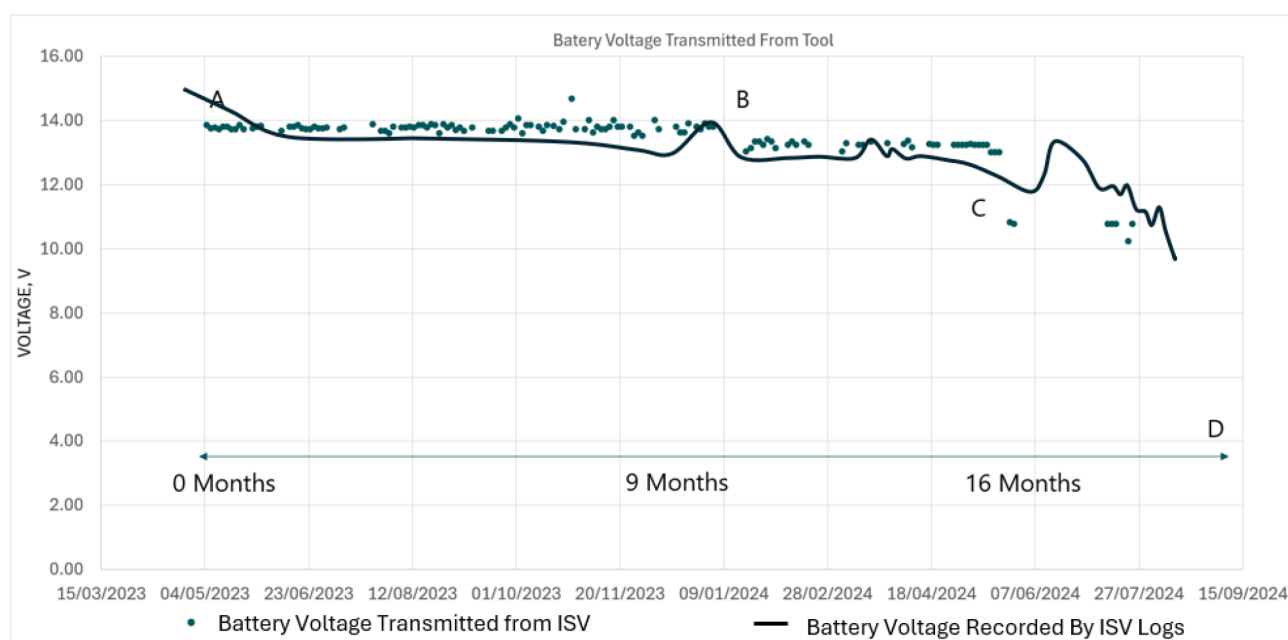


Figure 5—Plot of Vitality Pulse (VP) offset received from ISV in North Sea application. The time at which the Vitality Pulse is sent from the tool is translated into a remaining battery life. This application was designed for 18 months operation. The decline in remaining battery life communicated from the tool aligns with the 18-month planned operation. A – First VP indicating battery at full capacity. B – First sign of shift in battery life after 9 months operation, 86% remaining capacity. C – Second shift in battery life, 54% remaining capacity. This point denotes a sudden and significant drop off in battery capacity which is aligned with lithium battery performance curves. It is at this point that the client should plan for intervention to ensure replacement tools are ready. D – Tool Re-dress / changeout intervention date.

A logic-based system utilises on board monitoring to establish the ISV health. This feature ensures that fundamental aspects of the tool's function are analysed. On board risk management is built in to ensure that primary operating components are assessed; sensors, power and onboard computer. Should the tool identify

an issue with sensors or reduced power it will move to a closed position. In addition, where well debris; sand, scale, hydrates could be compromising the tool's ability to close and seal, it will activate an "unstick protocol" to remove any contaminants and ensure that in the event of detecting a trigger, it can close within its designed API 14A requirements.

All response event, open commands and vitality pulse frequency features are configurable, and are design to align with well conditions and operator needs.

Field Applications - Intelligent Safety Valve

We report five retrofit deployments in 3 wells of a wireless Intelligent Safety Valve (ISV) across platform and subsea offshore assets. For each case we summarise context, configuration, commissioning, and assess the long-term performance. Triggers referenced are LoSC (Loss of Surface Containment), ESD (Emergency Shut-Down), and Absolute (static threshold).

North Sea Platform (Well 1, Run 1)

Background. From 2009 to 2020, a North Sea operator had been required to make repairs to a TRSSSV on 7 separate occasions. As is common in aging assets, these interventions included the use of wire brush runs, scale dissolver, "flow tube exercise tools" and "hold open sleeves". In 2021, the hold open sleeve became stuck across the existing TRSSSV which rendered the valve inoperable and stuck in the open position. The obstruction across the valve prevented the remedial option of installing a traditional insert safety valve. Without a functioning safety valve, the well was shut in. This gas producer had a locked potential of 6MMscf/day.

The operator had the following requirements from a safety valve to reinstate production:

- Close and effectively shut-in production in a loss of surface containment (LoSC) event.
- Close and effectively shut-in production in an emergency shutdown event (ESD).
- Isolate flow and remain closed until intentional intervention is carried out.
- Demonstrates compliance with the applicable criteria in API RP 14B.
- Demonstrates compliance with the applicable design criteria in API spec 14A.
- Open on demand.

With its ability to respond to surface activated ESD events as well as LoSC and opening demands, an Intelligent Safety Valve (ISV) was selected as a solution to reinstate safety valve protection.

Design. Prior to installation, the ISV was configured to recognise the boundaries of a safe flowing condition. This was done using existing historic data to build a well model using nodal analysis. Along with wellhead pressure data trends, the nodal analysis production simulator model was used to quantify the pressure changes associated with both ESD and Loss of Containment scenarios.

The model was also used to determine the optimum choke position for generating a Vitality Pulse. VP target was to generate a pulse of sufficient magnitude to be detected in the flowing well head pressure FWHP, without causing flow disruption.

The predicted battery life was calculated based on the lithium cell type, temperature, number of batteries, idle power consumption and active power consumption for the forecast number of open/close cycles and VPs.

Installation and commissioning. Using standard intervention equipment, the ISV was installed and set on a packer using slickline in under 3 hours. Following installation the ISV was commissioned by modulating surface pressure and flow to replicate ESD or LoSC scenarios. When the ISV recognised a

pressure condition outside the boundaries of its programmed, safe flowing condition window, it closed and made the well safe. After the trigger and closure were confirmed, the tool was commanded to reopen from surface using a wireless protocol. All triggers and VP were confirmed to be functioning properly during the ISV commissioning.

Downhole Operation. The valve was operational for 17 months and 18 days. During this time, the valve was extensively tested by the operator for simulated ESD and LoSC events. The valve transmitted daily VPs which confirmed continued functionality of all its essential components while also communicating data to allow an estimation of remaining battery life. Battery life was communicated by a VP offset method. plots the VP offset timing in terms of battery voltage which is used as an indicator of remaining battery capacity. Battery depletion was in line with expectations. VP communications confirmed this, giving accurate, regular updates of the battery state while the tool was downhole. This information was used to predict remaining life and plan for tool replacement months in advance. After 535 days of operation, the tool reached its preconfigured low battery limit and permanently moved into the closed position. This low battery fail-safe closed position includes a safety feature that does not allow the operator to reopen the tool by command ensuring the well remains safe. After tool closure, the tool continued to monitor pressure, temperature and tool diagnostics. On this installation, the valve facilitated production at optimal rates for the well. It opened and closed on 20 occasions and transmitted 530 vitality pulses.

Retrieval and Installation of Replacement (Run 2). As a result of the real-time update on tool status, the well intervention for retrieval was planned to coincide with intervention on a neighboring well. The ISV was recovered to surface and logs from the valve's onboard memory were downloaded onsite and analysed. This logged data was used to optimize the configuration of the replacement ISV. Changes were made to optimise the new valve's response to recognition of ESD events and VP frequency was reduced to once every two days. The combined slickline intervention time for recovery and installation was under 7 hours. The newly installed valve continues to send regular vitality pulses and has a predicted battery life of over 3 years.

North Sea Platform (Well 2, Run 1)

Background. A North Sea operator had a cyclic gas-producing well with a control line leak which both rendered the Tubing Retrievable Sub-Surface Safety Valve (TRSSSV) inoperable and gave no option to install an insert safety valve. Due to the cyclic nature of production; traditional Sub-Surface Controlled Sub-Surface Safety Valves (SSCSSV) or velocity valve were not an option. The well had been shut in for 2 years and the operator required a safe, reliable solution that was not dependent on a control line. The ISV was selected to remediate this issue and restore production. It was planned for installation via slickline just above the existing TRSSSV using a retrievable packer. This would enable the operator to return the well back to production safely without replacing the upper completion. Particular attention was paid to the cyclic production strategy implemented by the client and to the risk of hydrates; the well operates on a 21-day build up, with production over a 7-day window. The well had a gas production potential of 5.5MMscf/day which had been locked in for 2 years.

Design. An important design consideration was the cycling of the well between flow and shut-in requiring frequent closing of the valve and, on startup, re-opening. The command protocol to re-open the valve was designed with this consideration in-mind. It should be a command that is simple for the operator to generate when starting up the well but must be sufficiently complex to ensure that the valve does not open unintentionally. The valve was configured to recognise a pressure drop of a specific magnitude over a defined time period, after a defined period of stabilisation, the valve recognised a 2nd drop in pressure again within the defined parameters, it would interpret this combined sequence as a deliberate command for it to open. The open command parameters were configured to allow the sequence to be easily achieved by the

operator with existing surface equipment. Nodal analysis modelling was used to define the safe operating window. Based on this modelling results, values for the ESD, LoSC and ABS triggers were determined.

Installation and Commissioning. The ISV was installed on standard slick line with a packer in under 2 hours. The valve was commissioned to verify correct function of the ESD, Loss of Surface Containment LOSC and Absolute ABS pressure triggers. Open commands were also tested after each closure.

Long-Term Performance. The ISV operated successfully for 2 years and 2 months. During this time, the valve performed 66 cycles to accommodate the cyclic production regime. The ISV was recovered and replaced earlier than designed to coincide with another ISV replacement planned for a neighboring well.

Retrieval and Installation of Replacement (Run 2). The ISV was retrieved and replaced with 6.5 hours of intervention time. On recovery, data was downloaded from the tool to show that the ISV had correctly responded to all events with over 66 cycles and 66 Vitality Pulses being transmitted. Log data from the retrieved ISV's onboard memory was analyzed to fine tune the opening command configuration for the replacement tool prior to re-installation. The re-installed tool is currently performing as expected and has an expected battery life of almost 4 years. Accurate battery voltage is being monitored through regular vitality pulse communication from the tool.

North Sea Subsea Application (Well 3)

Background. An operator in the North Sea was unable to function the existing DHSV in their subsea well due to a blocked control line. The issue could not be resolved and as a result the well was shut-in for over 10 years. The field had been on water flood and then blow down making the production regime and phase type from this well highly uncertain and very likely transient. Unlocking production from the well was a complex challenge. Any sub-surface safety valve solution must accommodate large fluctuations in produced fluid type (oil, water and gas), pressure and the associated temperature swings. As this was a subsea wellhead, a key requirement was to avoid frequent, costly interventions. The well had a production potential of around 7100bpd. The ISV was selected to remediate this issue and restore production.

Design. It was anticipated that going forward, the challenge of transient flow behaviour during well startup would exist after any time the well was shut-in. To mitigate this, the ISV was configured, for a defined period after opening, to ignore pressure trigger events during this startup phase. The startup period was configured specifically to account for the anticipated startup flowing conditions of the well, with the goal of allowing sufficient time to offload water and achieve a stable flow, while also ensuring the re-activation of the valve's trigger monitoring as required by the operator. This avoided the risk of the valve closing prematurely as a result of the transient behavior known to occur during well start-up. In addition, the valve was configured with an absolute trigger only. VPs were configured weekly to minimize disturbance of well production and maximize battery life.

Installation and Commissioning. The existing safety valve was locked open using a hold open sleeve. The ISV was installed in under six hours via slickline from a light well intervention vessel and set above the existing safety valve using a retrievable packer. Startup time of the ISV was set in the future in order to allow time for the well to be flowed, and water offloaded, prior to the ISV's closure trigger becoming active. On achieving a stable flow, and the ISV trigger becoming active, gas lift rates were adjusted to reduce the pressure at valve setting depth to a level where the ISV triggered and shut-in the well. This test, and the open command sequence that followed, were successful in proving valve functionality.

Long-Term Performance. The ISV has been operating normally in the well for the last year sending vitality pulses to confirm its function. The tool has successfully performed 10 open and closure cycles.

Results

In this section we summarize the results of the 5 deployments and 3 wells introduced in the previous section. [Table 1](#) summarizes the well data, production conditions for each well and justification for running the ISV technology. This data highlights the varied conditions for design, deployment, commissioning and operation of the ISV.

Table 1—Summary of Well Data

	North Sea Platform (Well 1)	North Sea Platform (Well 2)	North Sea Subsea (Well 3)
Environment	Offshore Platform	Offshore Platform	Subsea
Well Fluid	Gas	Gas Condensate	Oil
Tool Setting Depth	470ft	565ft	290ft
Flowing Temperature - tool depth	80°C	60°C	105°C
Flowing Pressure - tool depth	510 psi	4,300 psi	2,300psi
Flowrate	6MMscf/day	5.6MMscf/day	7,100bpd
Flow Regime	Stable	Cyclic	Liquid Loading
Lift	Natural	Natural	Gas Lift
Justification for Deploying Technology	Mechanical obstruction of DHSV	Leak in existing DHSV control line	Control Line Blockage
Well Specifics	Natural	Cyclic Well	High Water Cut & Gas Lift
Deployment Method	Slickline	Slickline	Slickline (LWI)
Hanging Device	Retrievable Packer	Retrievable Packer	Retrievable Packer
Tubing Size (in.)	5.5	5.5	5.5
ISV Size OD (in.)	2.5	2.5	2.5

[Table 2](#) summarizes the ISV configuration and functional performance parameters to enable a fair comparison of planned function versus actual for a long-term performance evaluation.

Table 2—ISV Configuration and Functional Parameters

	North Sea Platform (Well 1) Run1	North Sea Platform (Well 1) Run2	North Sea Platform (Well 2) Run1	North Sea Platform (Well 2) Run2	North Sea Subsea (Well 3)
Installation Date	Mid 2023	Early 2025	Early 2023	Early 2025	Mid 2024
Time in Well (yrs)	1.47	0.31 ₄	2.08 ₃	0.31 ₄	1.00 ₄
Battery Prediction (yrs)	1.5	3.3	2.2	3.8	2.9
# of Batteries	3	3	3	3	3
Triggers Active	LoSC, ESD, Absolute	LoSC, ESD, Absolute	LoSC, ESD, Absolute	LoSC, ESD, Absolute	Absolute
Communication Frequency	Daily Vitality Pulses	Bi-Daily Vitality Pulses	Daily Vitality Pulses	Bi-Daily Vitality Pulses	Weekly Vitality Pulses
Current Operation	Recovered for Redress	Live, Bi-Daily Vitality Pulses Received	Recovered for Redress	Live, Bi-Daily Vitality Pulses Received	Live, Weekly Vitality Pulses Received
Estimated Battery (yrs)	1.5	3.3	2.2	3.8	2.2
Operating Time (yrs)	1.46	0.31 ₄	2.08 ₃	0.31 ₄	1.00 ₄
Vitality Pulses	530	Regular Bi-Daily ₂	66	Regular Bi-Daily ₂	Regular, Weekly ₂

	North Sea Platform (Well 1) Run1	North Sea Platform (Well 1) Run2	North Sea Platform (Well 2) Run1	North Sea Platform (Well 2) Run2	North Sea Subsea (Well 3)
Active Triggers	ESD, LoSC, ABS	ESD, LoSC, ABS	ESD, LoSC, ABS	ESD, LoSC, ABS	ABS
Actual Cycles (Open or Close)	20	5 ₂	66	10 ₂	5 ₂
Intervention Time (hh:mm)	Install: 02:30	Recovery: 3:40 Install: 02:45	Install: 01:30	Recovery:03:30 Install:02:55	Install: 05:40
Method of Deployment	Slickline / Packer	Slickline / Packer	Slickline / Packer	Slickline / Packer	LWI / Slickline / Packer
Surface Equipment	Existing WHPG	Existing WHPG	Existing WHPG	Existing WHPG	Existing WHPG

Notes:

- ¹ From tool pickup to running tool recovery.
- ² Tools are currently live downhole.
- ³ Recovered early to coincide with intervention plans.
- ⁴ Correct as of 19th August 2025.

Discussion

Long-term Performance

In all five cases the ISV's performance met the operators' requirements of reinstating safe, long term production where existing safety valves were inoperable. compares the ISV predicted life with the actual life. Well 1, Run 1 and Well 2, Run 1 both achieved their predicted lives. The remaining examples are still operating in line with predicted battery life. Well 1, Run 1 battery triggered a low battery safe mode which moved the tool to a permanent closed position in line with the tool configuration. With regular VPs from this well the operator was forewarned that the tool may move to a safe, closed position and was able to plan an intervention well in advance for a timely replacement.

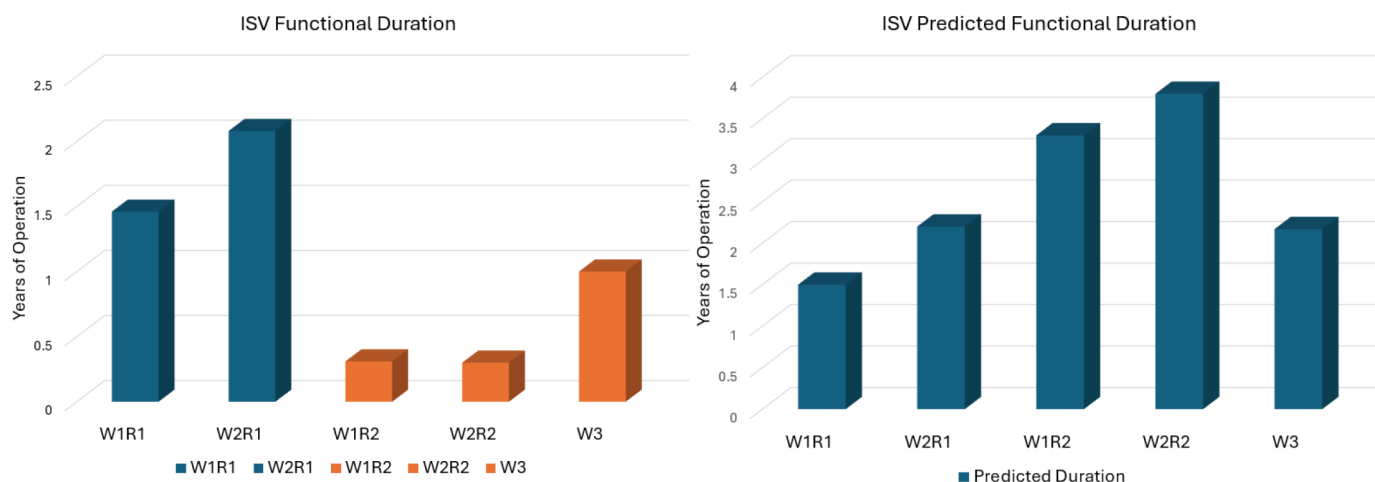


Figure 6—Duration of the ISVs in hole. W1R1 and W2R1 lasted the predicted time. W1R2, W2R2 and W3 are currently live downhole.

Four of the cases were platform installations where the original ISVs were recovered after 1.5 and 2 years operation and new tools installed. New generation tools with improved power management and battery technology increased the battery life predictions up to 3.3 and 3.8 years for Run 2. Although using the same battery and power optimization technology as wells 1 and 2; the subsea case has a lower predicted battery

life of 2.2 years which is a result of the higher flowing temperature anticipated in the late life of well as the production phase changes.

In the four cases with dynamic triggers activated, the ISV successfully adjusted the boundaries of its safe working pressure window which allowed for the operator to produce the well at the optimal rate without risk of early inadvertent closure when the well's flowing pressure dropped to a certain value. This dynamic triggering capability has removed the requirement for frequent interventions to reset and recalibrate traditional velocity type valves and, as a result, greatly extends the functional life of a safety valve downhole.

Intervention

Intervention times are taken from the time the ISV is picked up from deck to the time the packer running tool is back to surface. They do not account for rig -up and testing of intervention equipment or any preparatory runs (plug recovery, brush, drift). The duration of all interventions was typical for a slickline deployed packer. The results for all scenarios consistently demonstrate that the ISV can be efficiently deployed and commissioned (Figure 7). No special equipment was required for these deployments.

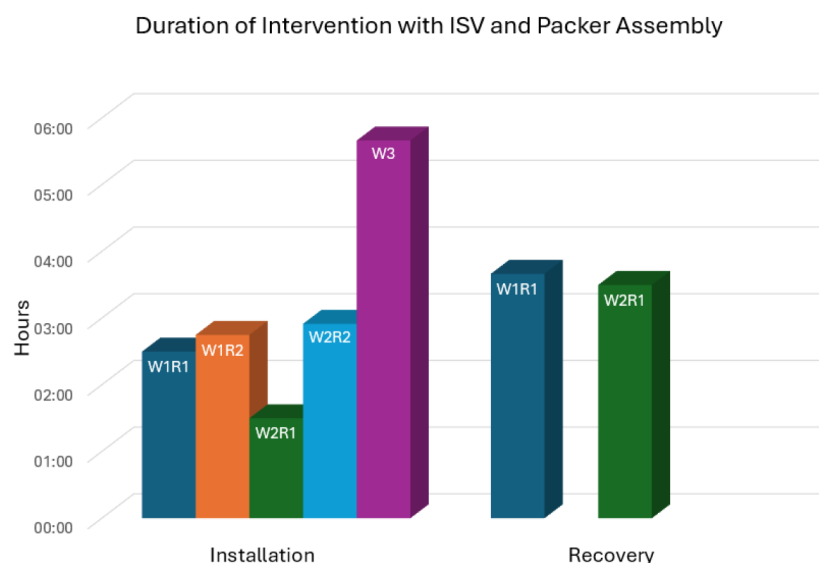


Figure 7—Duration of intervention time. Times are from the lifting of the ISV from deck to the return of the packer running tools to surface.

Triggers

All trigger mechanisms operated as per design during commissioning and long-term operation across the five cases. Four cases used both dynamic and absolute pressure triggers to monitor the safe working window of the wells. For the cases where the ISV was run in the same well for a 2nd time, the configurations were marginally changed between R1 and R2 to optimize valve operation for the well's predicted performance. In other words, the ESD triggers were adjusted to accommodate the reduced rate of change of pressure in an ESD scenario with a depleted well.

Vitality Pulses

The ISVs all transmitted VPs to surface on a regular basis as designed. This confirmed that the valve movement was unobstructed and all critical systems were functioning normally. The VPs also send daily information on battery health. During Run1 in Well 1, the VPs provided timely indication of depleting battery voltage which allowed sufficient time to plan intervention and ISV replacement. The VP detection and monitoring used the operator's existing surface sensors. The number of detected VPs from Well 1 are significantly higher than that of Well 2, due to Well 2 being a cycling producer and therefore not flowing for the majority of the time and hence not transmitting VPs when shut-in. The valves that are currently

live downhole are expected to continue to transmit VPs for their predicted operating duration. In flowing conditions, the VPs have been successfully detected in the WHP data to confirm the valve's status and battery voltage. There are currently no other safety valves in the industry with this level of monitoring. In the case of Wells 1 and 2, the maximum time to recognise a non-functioning valve is reduced to between 24 and 48 hours depending on the VP configuration. In the case of all other valves on the market, this time could be as high as 3-6 months.

Wireless Command of Open / Close Cycles

As Well 2 is a cyclic producer, the ISV was closed each time production ceased and an open command was sent to the valve ahead of production startup. For this reason, during Well 2, Run 1 logged 66 occasions where it was cycled. This is significantly higher than Well 1, Run 1 which was cycled only 20 times for testing and planned maintenance. were logged. In all cases, open commands to the valve have been successful and easy for the operator to perform. There were no cases where an ISV was opened unintentionally.

Unlocking Trapped Potential

The actual production data was not available for this paper, so an estimate of unlocked production has been calculated based on the average well production for UKCS in 2023 and the cumulative production days with the ISV installed. For the wells and deployments reviewed in this paper a total unlocked production of 811kBOE is estimated over a 3.4 year cumulative production period.

Environmental

There are a variety of retrofittable solutions on the market to remediate failed safety valves, [Abbott et al 2024](#) outlines provided a CO₂ emission comparison of a storm choke vs the wireless Intelligent Safety Valve (ISV) over a long duration. Intervention and production downtime were fixed for both applications. The number of interventions was a key differentiator resulting in a 75% reduction in CO₂ for the ISV deployment. It is assumed that the same tool is used throughout the four-year period, with intervention for re-dress only. In addition to CO₂ reduction there is a substantial benefit gained by extending the useful life of the asset and producing the locked in hydrocarbons. The versatility of this technology stems from the ability to retrofit, autonomous response capable of adapting to a wide variety of production conditions and the wireless command and control removing the need for cables or control lines for actuation.

Conclusions

In this paper we have reviewed in detail a wide variety of case histories where the wireless ISV technology has been deployed to remediate failed safety valves and return shut-in wells to production. We have demonstrated the versatility of the ISV in different environments, well types and under a range of downhole and production conditions. The wireless command, control and autonomous function have been proven during installation, commissioning and long-term operation. Single and multiple safety triggers have proved successful under regular testing. The vitality pulse has functioned as designed, actively testing tool function while communicating battery status and tool health to surface. The analysis of deployment data shows both an accurate forecast of tool life and improvements in battery technology and power management that are helping to extend the downhole life in the latest generations of ISV. The benefits of ISV technology are evident: extending the life of existing assets by returning shut-in wells to production while minimising workover cost and environmental impact.

Nomenclature

ABS — Absolute Trigger

API — American Petroleum Institute

CO ₂	— Carbon dioxide
DH	— Downhole
DHSV	— Down Hole Safety Valves
ESD	— Emergency Shut-Down
ISV	— Intelligent Safety Valve
ICV	— Intelligent Control Valve
LoSC	— Loss of Surface Containment
LWI	— Light Well Intervention vessel
R1	— Run 1
R2	— Run 2
TRSSV	— Tubing Retrievable Subsurface Safety Valve
VP	— Vitality Pulse
UKCS	— United Kingdom Continental Shelf
W1	— Well 1
W2	— Well 2
W3	— Well 3
WG	— Wireless Gauge
WHPG	— Surface Wellhead Pressure Gauge
WRSSSV	— Wireline Retrievable Sub-Surface Safety Valve
yrs	— Years

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